

of price, quality and support—all to the benefit of the user. Even branded PCs are including scanners in their packages to boost sales, as increasingly more home users have begun to opt for this device. A scanner with a resolution of 1200 dpi and 48-bit colour depth can be bought for, as low as, Rs 3,500.

Our comparison test rounds up more than 20 scanners available in the market to offer you the best suited one for your varied tasks.

How we tested

Test bed used: Prior to running the tests on every scanner, a ghost image of Windows XP is loaded, after which, the necessary drivers are also loaded. The desktop resolution is kept at 1024 x 768 x 16 at a refresh rate of 75 Hz. The test machine used is P4 2.0 GHz with 256 MB RD RAM and 40 GB ATA/100 hard drive on an i850E motherboard. The monitor used is Samsung Samtron 75E. Windows XP with SP1 is loaded on the System with only Photoshop 7.0 and Office XP.

Each scanner was evaluated for speed and quality apart from noting relevant features. The tests include:

Speed test: Here we logged the time taken by the scanner to preview and scan documents, and our test image at 300 dpi and 1200 dpi resolutions (line art and colour). This test was given the greatest importance for the home user segment.

Image quality test using an IT8 card: An IT8 card is used by professionals the world over to determine the colour purity of a scanned image. This card is scanned at 300 dpi and 1200 dpi optical resolutions to check out the colour reproduction in light and dark region. This test is of importance to professionals who demand colour accuracy from the scanned document.

Image quality test using a resolution chart: In this test, a resolution chart comprising lines is scanned at 300 dpi and 1200 dpi optical resolution; then, using the magic wand feature of Photoshop, all the scanned blocks are analyzed. The scanner which can differentiate between scanned lines in all the five blocks is awarded the maximum marks. This test helps in determining which scanner is best in picking up minor details from

a medium to be scanned.

Image quality test using a photograph: In this test, a highly detailed image is scanned and analysed with regards to the reproduction of the details and colours. The test image was divided into different areas, which showcased different qualities of a scanner and marks were assigned accordingly.

OCR test: This test is important because it determines the ability of a scanner to scan a text document accurately such that it can be used by an OCR-software for extracting the text therein. It is vital for a scanner to scan the image precisely for the OCR software to give accurate results. The medium is scanned at the recommended resolution of 200 dpi.

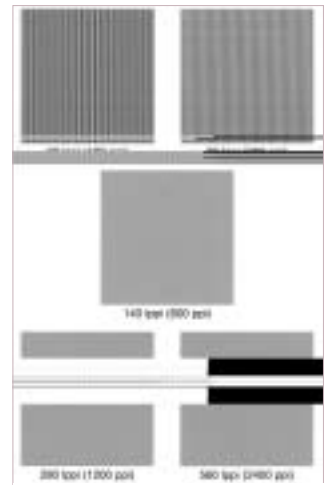
Features: This is one of the main ingredients of decision-making for scanners. We, therefore, noted down as many as 12 different features from which to draw a clear picture of a particular device. We also noted down special features such as Automatic Document Feeder and Negative Scanning attachment. An appropriate weightage was assigned to the different features, logged according to their importance in their respective segments.

After gathering the relevant scores from all the tests performed, we calculated the Suitability to Task Index (STTI). This STTI score helps us determine which scanner is best suited for a particular application area, hence making the buying decision much simpler for you. The greater the STTI score, the better the scanner. To further simplify the whole process, price is also factored in to give a single score that reflects the overall buying worth of a scanner.

How they fared

A 1200 dpi resolution has become the bare minimum for even the entry-level scanners. The entire range we tested—except a few, such as the Umax AstraSlim 600, which supported 36-bit colour depth only—supported this hardware resolution and a 48-bit colour depth.

Each of the top five scanner brands that participated in this comparison test had something unique to offer. Some, such as the HP 2300c, were very cheap and still managed to deliver quality. We also saw USB 2.0 finally making its way in to the scanner segment, but the performance numbers did not reflect the



Resolution chart used to determine scanner capability to pick-up finer details



We chose this image to determine overall performance of the scanners, and looked for the quality of colour reproduction and minute details

Determines the scanners capability to detect dark and light regions

The minute color deviation in neighbouring boxes really tests the scanners



Tests the scanners ability in picking tonal deviation